Review：

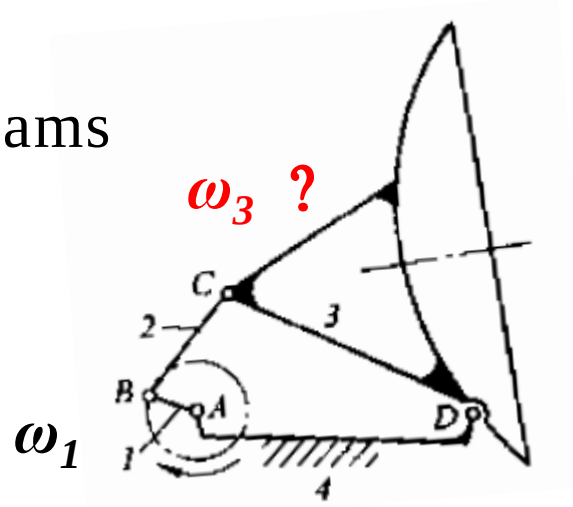
1. Drawing of Their Kinematic Diagrams

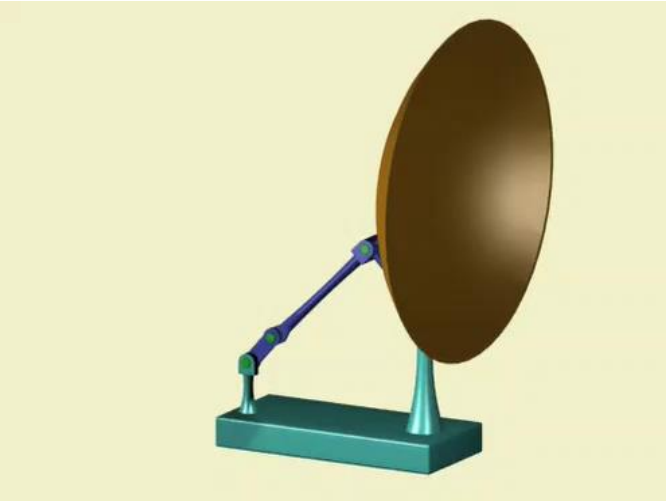
   Easy for mechanism analysis

2. Degree-of-Freedom Calculation

   Can it move ?

Today： $\omega_1 \rightarrow \omega_3$?





Radar Mechanism

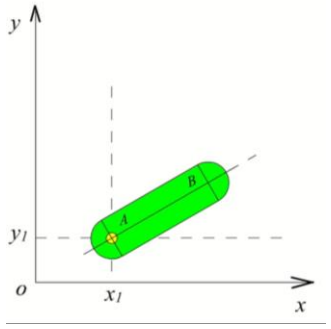
For a planar rigid body, at a given instant, there is a point with zero velocity.
This point is the absolute instantaneous center .

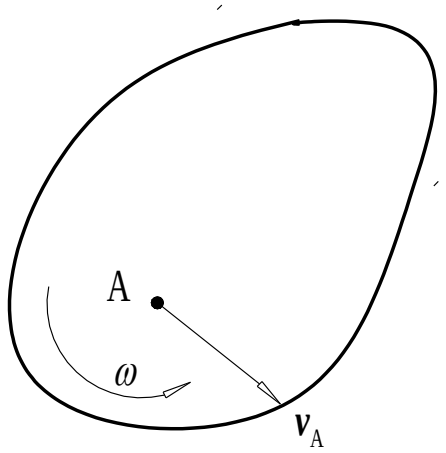
If point A has velocity v_A, and the angular velocity is ω, where is the instantaneous center?

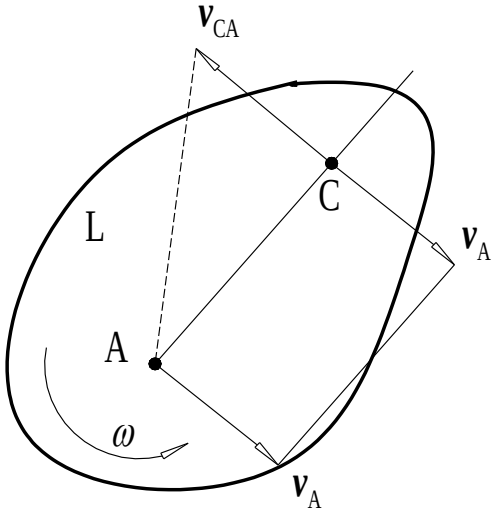
$$\boldsymbol{v}_C = \boldsymbol{v}_A + \boldsymbol{v}_{CA}$$

$$v_C = v_A - \omega \times \overline{AC}$$

$$\overline{AC} = \frac{v_A}{\omega}$$



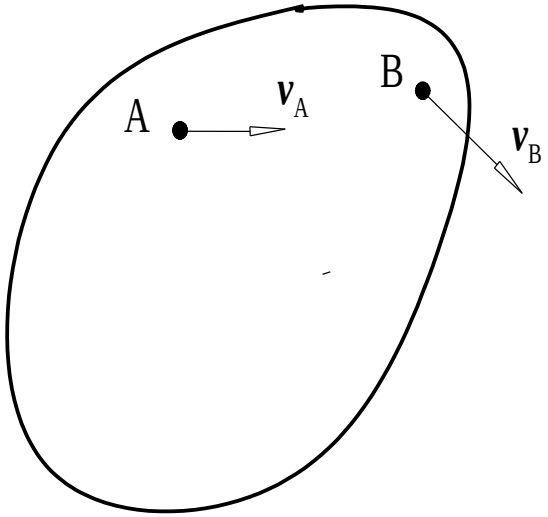


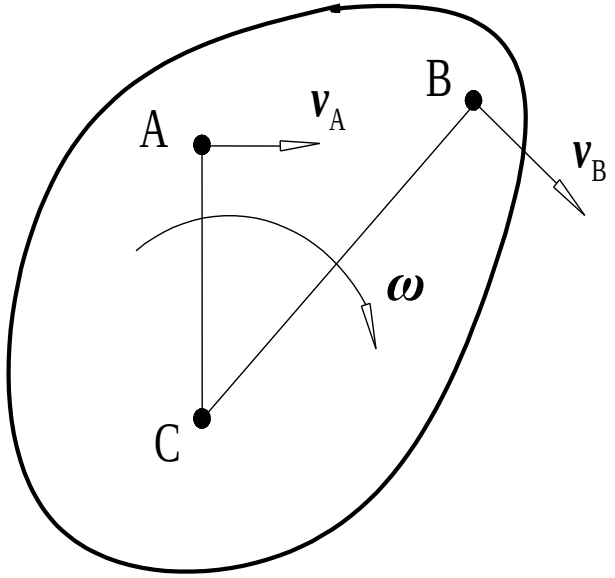


The instantaneous center is not necessarily a physical pin.
It may be outside the link.
It may even move from instant to instant.
But at this instant, it acts like the rotation center. $v_A = \omega \overline{AC}$

Find the IC from Two Velocity Directions

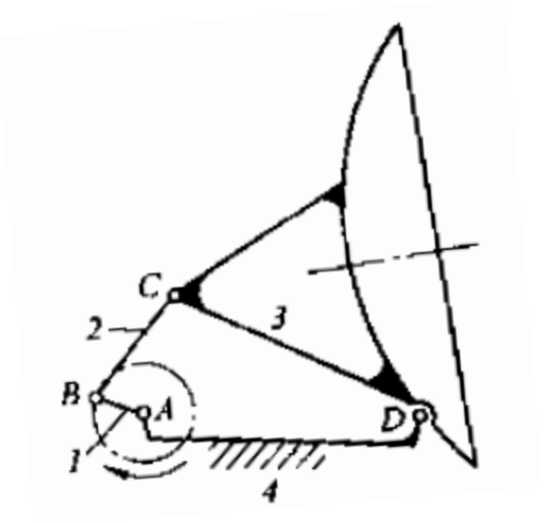
If we only know the velocity directions of points A and B, can we locate the IC?

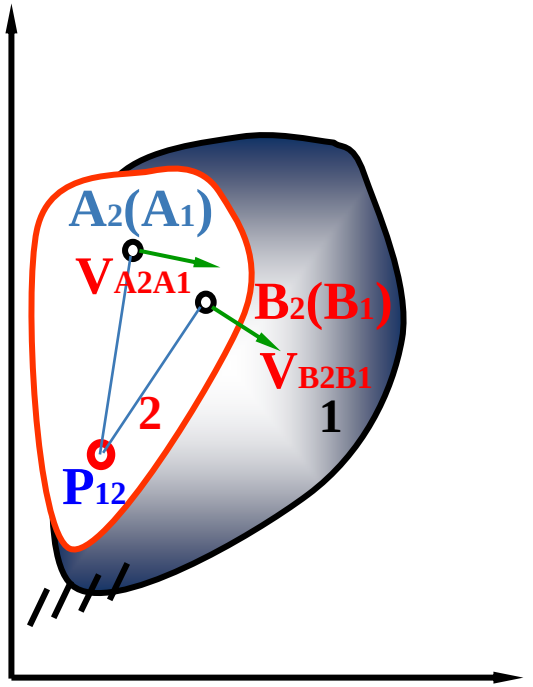




Relative Instantaneous Center

For two links i and j , they have a relative instantaneous center P_{ij} with the same velocity (relative velocity is zero)

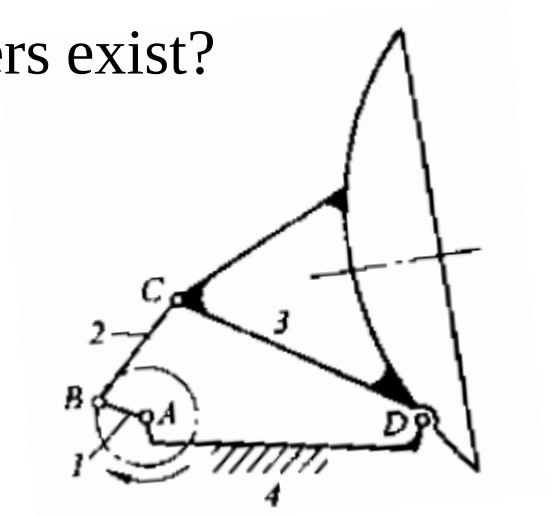




For two links, the instantaneous center is not always a zero-velocity point.
If one link is the ground, then it is an absolute IC.
If both links move, it is a relative IC.

If a mechanism has N links, how many instantaneous centers exist?

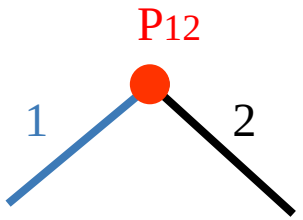
Every pair of links has one instantaneous center.
So this is a combination problem.

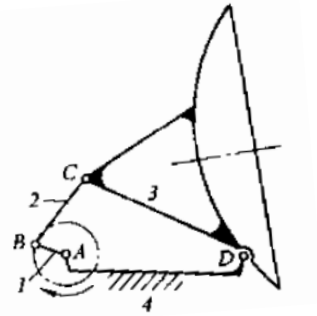


$$K = C_N^2 = \frac{N!}{2! \times (N-2)!} = \frac{N(N-1)}{2}$$

ICs of Directly Connected Links

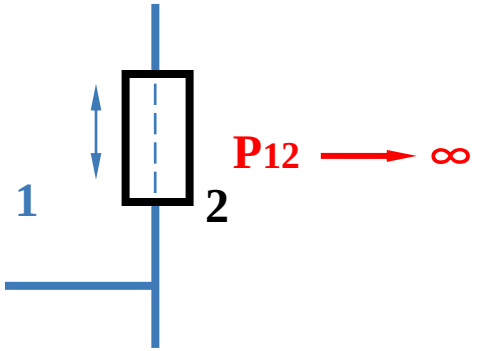
Where is the IC if two links are connected by a revolute joint?

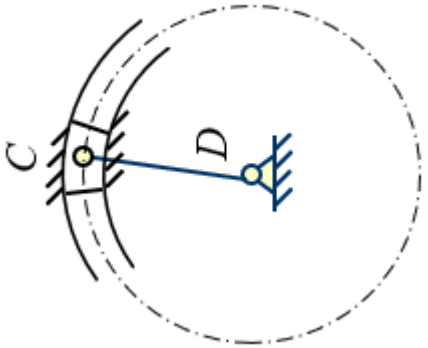




At the center of the revolute joint.

Where is the IC if two links are connected by a prismatic joint?





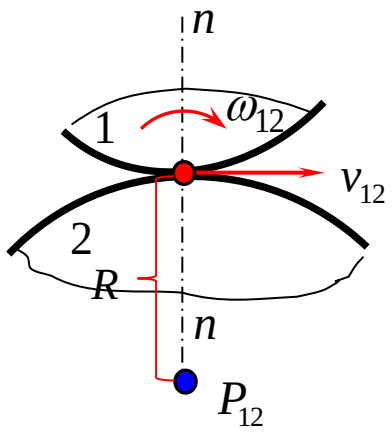
At infinity, perpendicular to the sliding direction.

For rolling contact, where is the IC?

At the contact point.

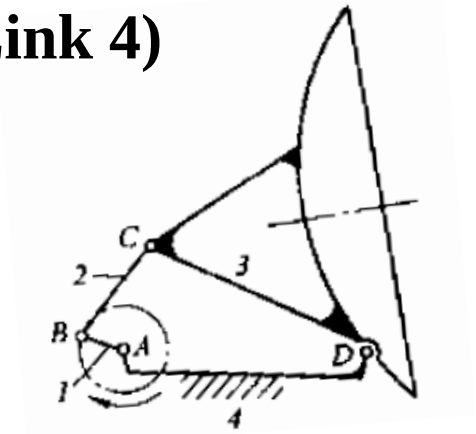
For rolling plus sliding contact, where is the IC?

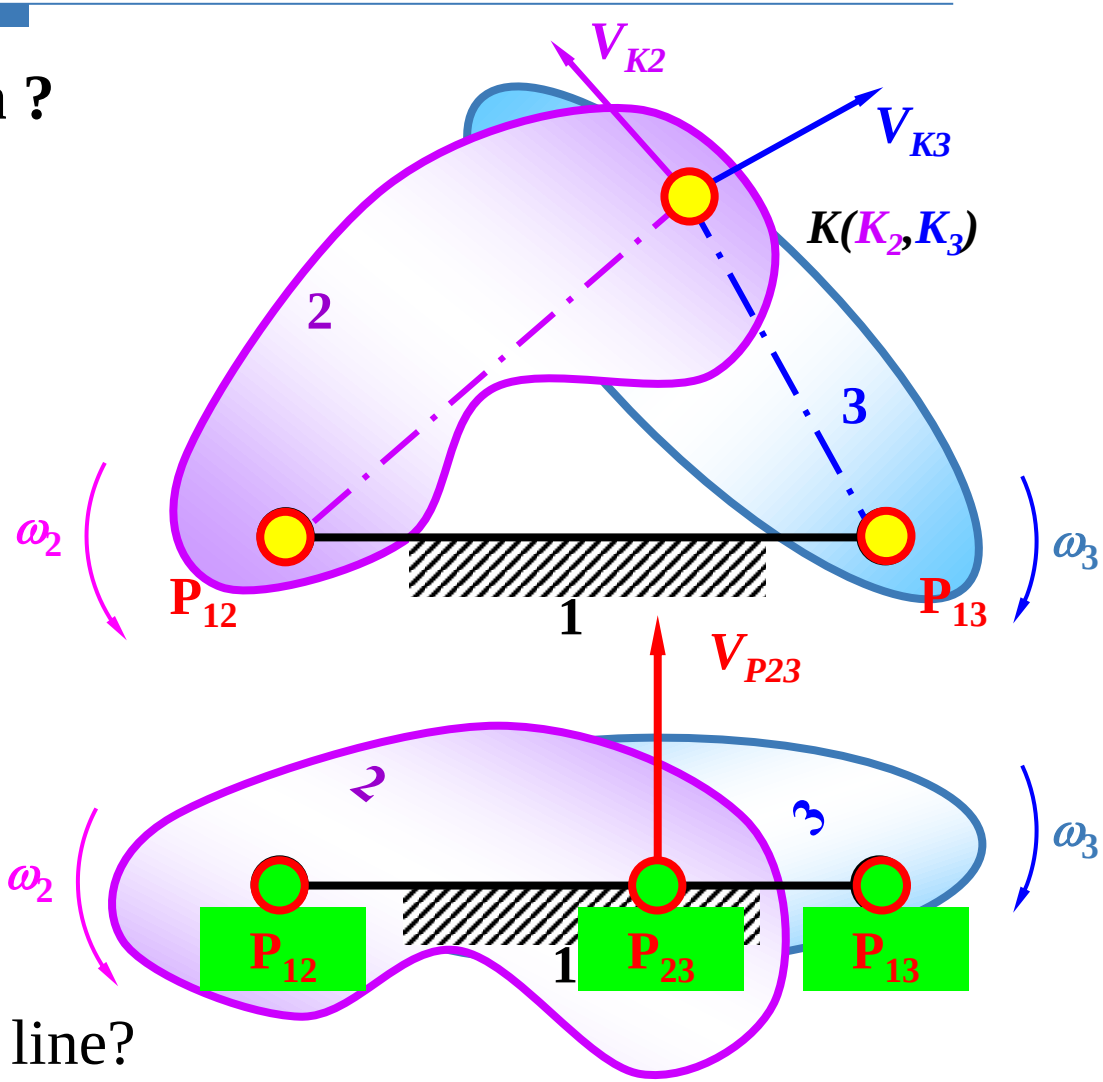
On the common normal through the contact point.



$$R = \frac{v_{12}}{\omega_{12}}$$

How about ICs of Links with no direct connection ?
(e.g., Link 2 and Link 4)





Kennedy's Theorem:
The three instantaneous centers of any three links must lie on one straight line.

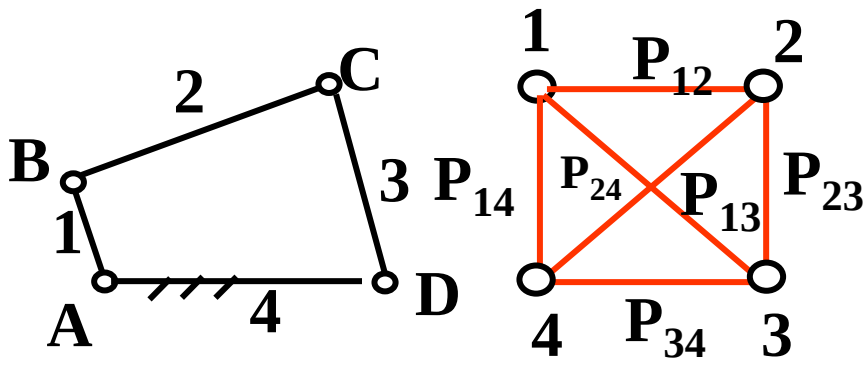
Prove?

What happens if the three ICs are not on one straight line?

The assumed equal-velocity point would give inconsistent velocity directions!

How to locate ICs ?

Instantaneous Center Polygon Method



Steps

1.Represent each link by a vertex.

2.Connect two vertices to represent one IC.

3.Use triangles to apply Kennedy's theorem.

4.Use common sides of triangles to locate unknown ICs.

This method helps us organize ICs.

It is like a map.

The mechanism gives us some ICs directly, and Kennedy's theorem helps us find the rest.

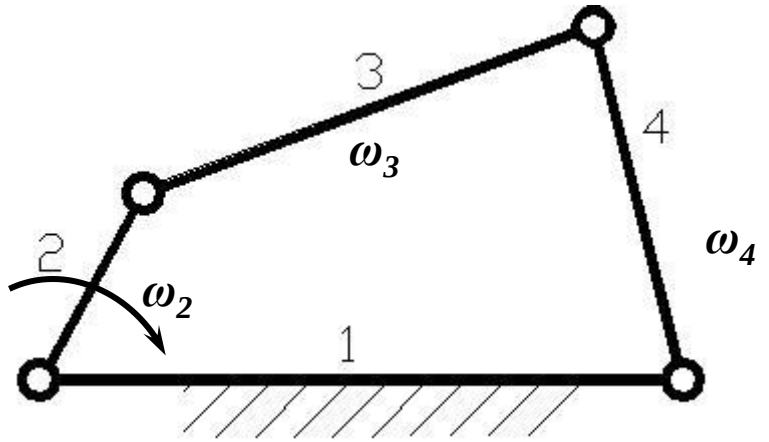
Problem:

For a four-bar mechanism:

1.How many ICs are there?

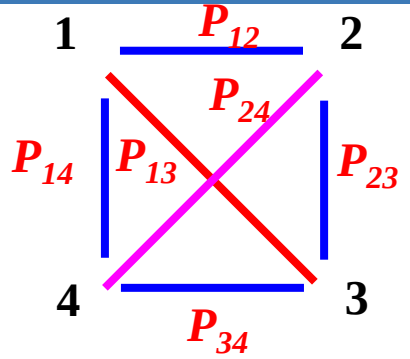
2.Where are they?

3.If ω_2 is known, how can we find ω_3 and ω_4 ?



$$K = N(N-1)/2$$

$$= 4(4-1)/2 = 6$$



2、Identify all ICs

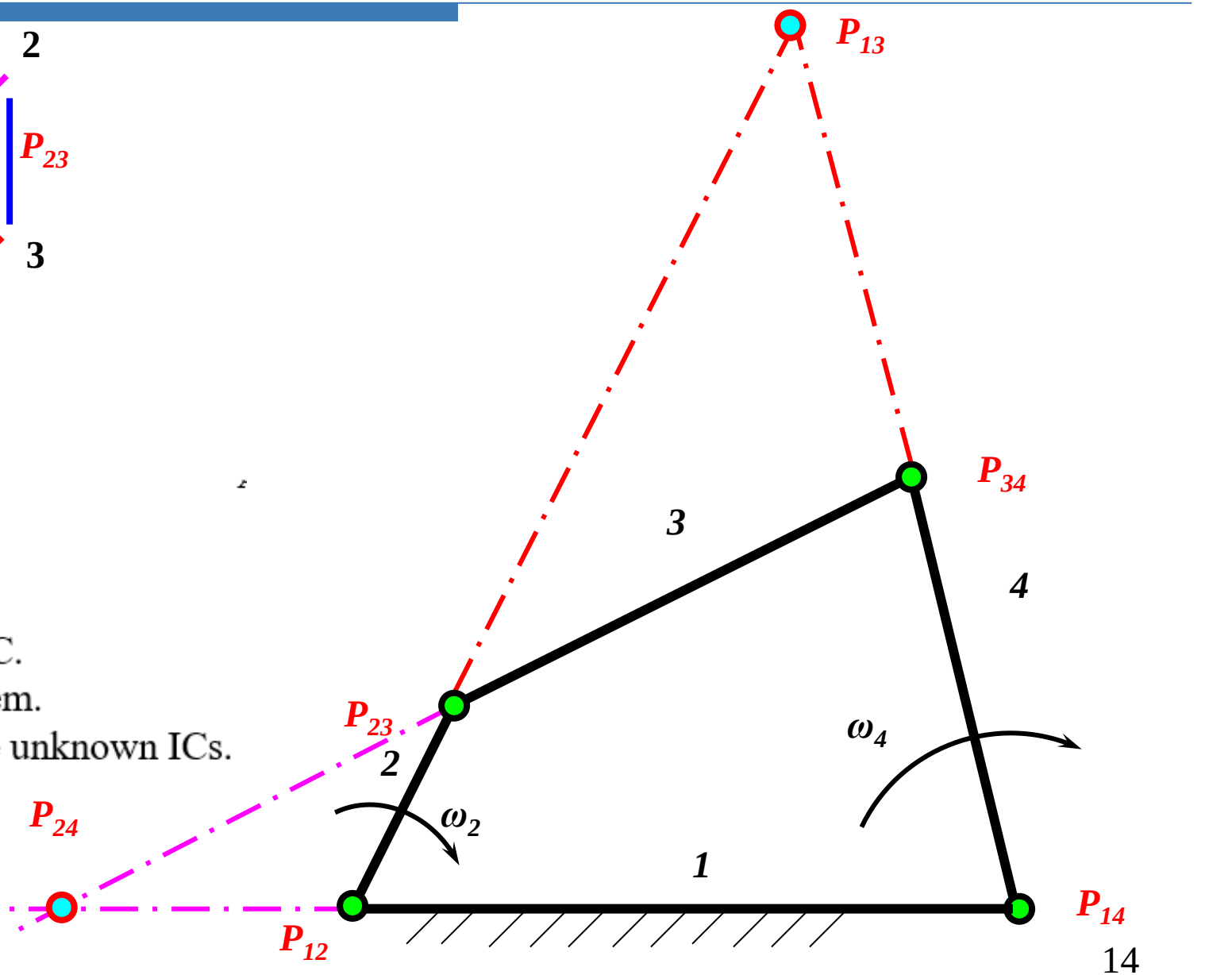
P_{13}? 1-3-4 and 1-3-2

P_{24}? 2-4-1 and 2-4-3

Instantaneous Center Polygon Method

1.Represent each link by a vertex.
2.Connect two vertices to represent one IC.
3.Use triangles to apply Kennedy's theorem.
4.Use common sides of triangles to locate unknown ICs.

The three instantaneous centers of any three links must lie on one straight line.



1.If ω_2 is known, how can we find ω_4 ?

Key principle:

At P_{24}, link 2 and link 4 have the same velocity.

$$v_{p24} = \omega_2 \overline{p_{12}p_{24}}$$

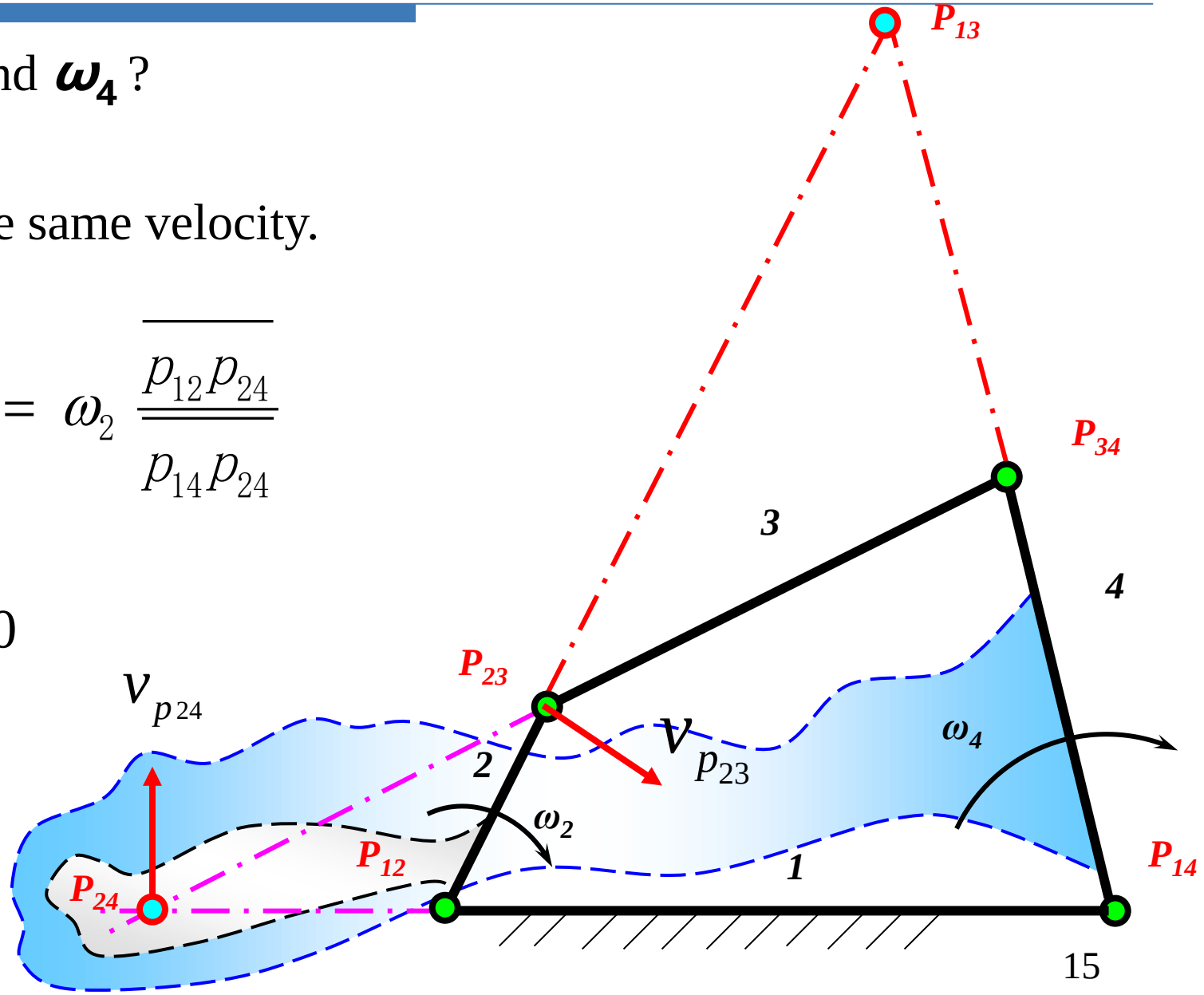
$$v_{p24} = \omega_4 \overline{p_{14}p_{24}}$$

$$\therefore \ \omega_4 = \omega_2 \frac{\overline{p_{12}p_{24}}}{\overline{p_{14}p_{24}}}$$

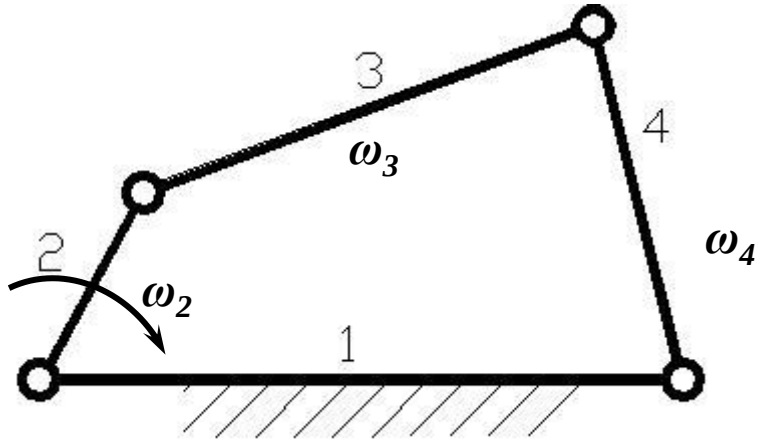
How about ω_3 ? $v_{p13} = v_1 = 0$

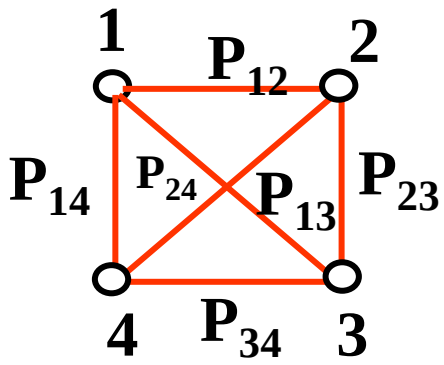
$$\therefore \ V_{p_{23}} = \omega_3 \overline{p_{13}p_{23}} = \omega_2 \overline{p_{12}p_{23}}$$

$$\therefore \ \omega_3 = \frac{\overline{P_{12}P_{23}}}{\overline{P_{13}P_{23}}} \omega_2$$



Summary of IC-Based Velocity Analysis





- Count the total ICs.
- Locate all ICs. (more exactly, locate the ICs between links with unknown velocities and links with known velocities)
- Use equal velocity at an IC to calculate unknown velocities.

Problem:

Given the crank angular velocity ω_1, find the slider velocity v_3.

Questions:

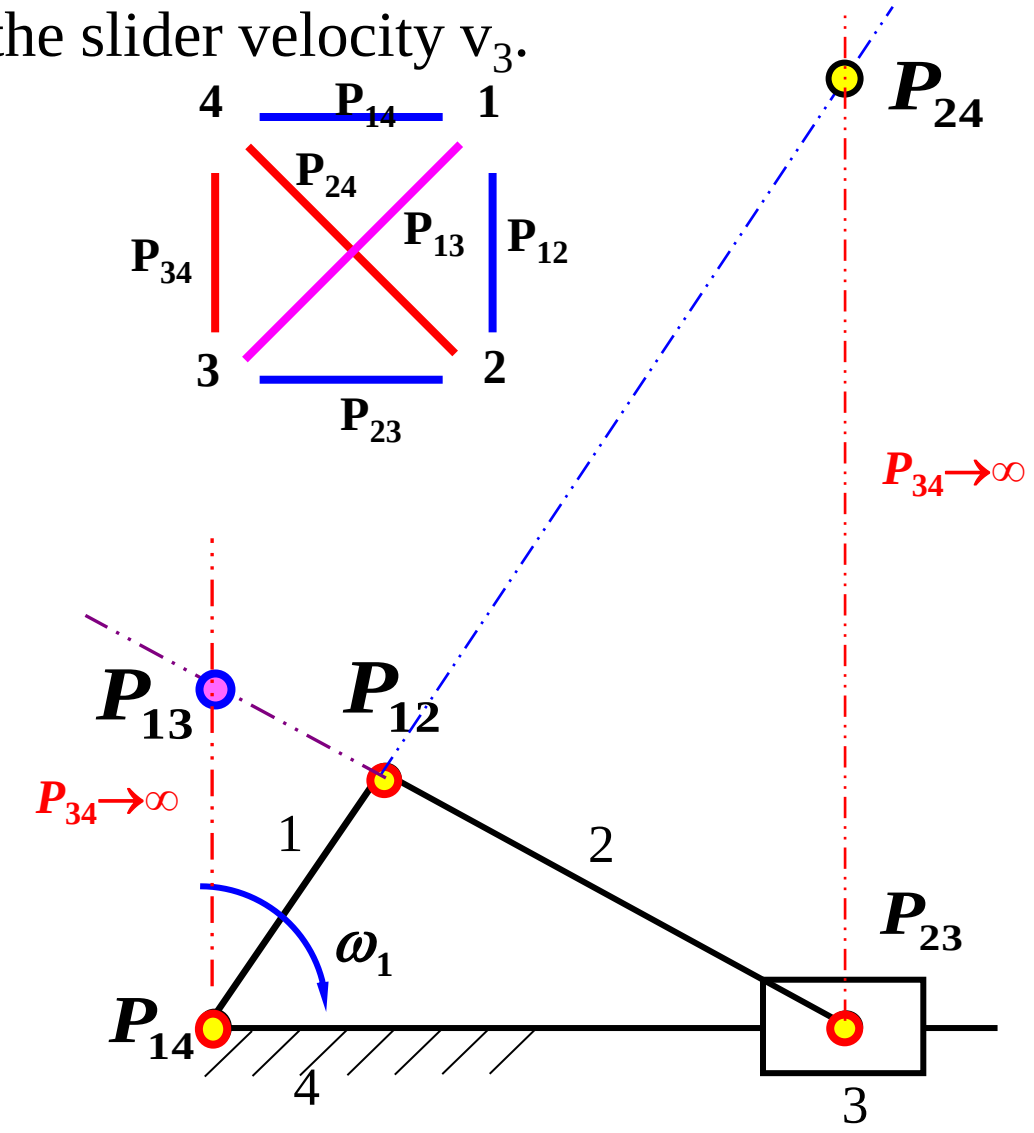
1.How many links are there?

2.How many ICs?

3.Which IC helps us find v_3?

$\mathbf{P_{24}}$?

$\mathbf{P_{13}}$?



Problem:

Given the crank angular velocity ω_1, find the slider velocity v_3.

Which IC helps us find v_3? $\mathbf{P_{13}}$?

Key principle:

At P_{13}, link 1 and link 3 have the same velocity.

$$v_{P_{13}} = \omega_1 \left(\overline{p_{14} p_{13}} \right)$$

$$v_3 = v_{P_{13}}$$

$$\therefore \; v_3 = \omega_1 \left(\overline{p_{14} p_{13}} \right)$$

